2758-T4 Advanced Topics in Machine Learning
Project Descriptions

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Outline

- Project description
- Evaluation criteria

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Final Project: Build an AI-Driven Startup

Philosophy: Business First, AI as the Engine

- **The Challenge:** Conceptualize and build a startup where AI is the core driver of value creation, not just a technical gimmick
- **Identify the Gap:** Start with a concrete business problem or market friction. What specific "pain point" are you solving?
- **AI-Enabled, not necessarily "AI-Only":** Your business model doesn't have to sell AI technology itself. It can be a traditional business model (e.g., legal service, retail, financial research) radically transformed by AI efficiency.
- **Both Carbon & Silicon Solutions are welcome:** Human-in-the-loop and fully Autonomous Agents workflows are equally welcome.

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The Deliverables

Deliverable 1: The Business Plan

- Target market, value proposition, and Go-to-Market (GTM) strategy.
- **GenAI Transparency Logs:** A detailed appendix showing exactly how you used AI tools (Claude, ChatGPT, etc.) during your ideation, coding, and writing processes.

Deliverable 2: Full-Stack Deployable Solution

- You must deliver a working prototype with both **UI (Frontend)** and **AI Backend**.
- No more command-line only tools. Leverage platforms like Firebase, Streamlit, or Project IDX to create a user-facing demo.
- We want to see the *step-by-step* execution, from prompt engineering to system integration.

AI Unit Economics & Financials

Cost Structure (The "Silicon" Cost):

- **Inference Costs:** Will you use API calls (e.g., Gemini 1.5 Pro, GPT-4o)? Calculate your token costs per user/transaction.
- **Hosting Costs:** Are you deploying local/open-source models (e.g., Llama 3) to save API costs? Factor in cloud GPU (AWS/GCP) hosting expenses.
- **Data Costs:** Vector database storage for your RAG pipeline.

Revenue Model:

- Demonstrate a clear path to profitability based on your AI Unit Economics.

Evaluation Criteria

- (1) **Commercial Innovation & Feasibility:** Does this solve a real problem? Is the financial model (ROI, Token Economics) sustainable?
- (2) **Technical Execution & Prototype:** Does the full-stack solution actually work? What is your tech stack?
- (3) **Defensibility & Safety:** What is your "Moat"? (Why can't OpenAI just build this as a feature?). How does your system handle AI hallucinations or data privacy?
- (4) **Presentation & Delivery:** Pitching your startup convincingly to investors.

The Judging Panel: Human + AI

A Dual-Review System

- **The Carbon Judges:** PhD-level staff and faculty will evaluate your business logic, market fit, and presentation skills.
- **The Silicon Judges (LLM-as-a-Judge):**
  - Your business plan and technical architecture will be fed into a specialized evaluation LLM.
  - The AI Judge will pressure-test your logic, simulate edge cases, and evaluate your cost structure.
  - *Warning:* The AI Judge is highly critical of "wrapper" startups with no real moat.
- **Academic Integrity:** Unacknowledged use of external resources or hidden GenAI usage will result in a warning, grade deduction, or zero.

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